Qualification Pack

Embedded Software Engineer

QP Code: NIE/ELE/Q0201

Version: 2.0

NSQF Level: 6

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Contents

NIE/ELE/Q0201: Embedded Software Engineer

Brief Job Description

The Certificate course is targeted for creating qualified professionals in the field of Embedded & IoT. Qualification has been developed in consultation with industry experts in the domain, aiming at Empowering the future workforce with necessary skills for employment and entrepreneur development of the qualifier.

Personal Attributes

The purpose is to develop the skill set required for the Design and Development of Embedded System Applications using suitable Hardware and Software tools. This course offers a range of topics of immediate relevance to the industry and makes the participants exactly suitable for the Embedded Industry

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. NIE/ELE/N0201: Embedded System Design with Embedded C Programming on ARM Cortex Micro-controllers
2. NIE/ELE/N0202: Implementation of Complex Embedded Systems using suitable Operating Systems
3. NIE/ELE/N0203: Real-Time Embedded System Design with RTOS
4. NIE/ELE/N0204: IoT Application Development with Python and Data Analytics
5. NIE/ELE/N0205: Complex Embedded Application Development with an understanding of Embedded Protocols and Device Driver Concepts
6. DGT/VSQ/N0104: Employability Skills (120 Hours)

Qualification Pack (QP) Parameters

Sector	Electronics
Sub-Sector	Sensor, Instrumentation and Cyber Physical System
Occupation	Embedded System Design, IoT Solution
Country	India
NSQF Level	6

Credits	32
Aligned to NCO/ISCO/ISIC Code	2021/ITES/NIELIT/04219
Minimum Educational Qualification & Experience	B.E./B.Tech (Final Year B.E./B. Tech in Electronics/ Electronics & Communication/ Electrical/ Electrical and Electronics/Instrumentation/ Electronics & Instrumentation / Instrumentation & Control /Biomedical /Computer Science/Information Technology or above) with 0-6 Months of experience OR M.Sc (Final Year M.Sc. (Electronics)) with 0-6 Months of experience OR B.Sc (Completed 3 Year B.Sc (Electronics)) with 1 Year of experience OR Diploma (Completed 3 Year Diploma (EEE/ECE/CS) after 10th) with 2 Years of experience
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	23/06/2026
NSQC Approval Date	23/06/2023
Version	2.0
Reference code on NQR	QG-06-EH-00567-2023-V2-NIELIT
NQR Version	2

NIE/ELE/N0201: Embedded System Design with Embedded C Programming on ARM Cortex Micro-controllers

Description

Gain expertise in embedded systems with a focus on ARM Cortex processors for deterministic real-time applications. Master memory management, pointers, and data structures. Understand ARM architecture, including pipeline structure and interrupts. Explore real-time operating systems for efficient task management. Apply concepts through hands-on projects for practical experience.

Scope

The scope covers the following :

- To set the required background in embedded system concepts, Embedded C language such as Memory management, Pointers, Data structures, and architecture of the ARM Cortex processor for highly deterministic real-time applications.

Elements and Performance Criteria

'C' and Embedded C

To be competent, the user/individual on the job must be able to:

PC1. Understand the fundamentals of C programming

PC2. Programming with embedded C in contrast with C language

PC3. Develop applications using Embedded C Programming

Embedded Concepts

To be competent, the user/individual on the job must be able to:

PC4. Understand the various hardware and software architecture of embedded systems

PC5. Hands-on exercises on development and debugging tools

Introduction to ARM Cortex Architecture

To be competent, the user/individual on the job must be able to:

PC6. Understand The ARM Architecture, Overview of Cortex Architecture, Register Set and Modes

PC7. Hands-on exposure to ARM Cortex M4 Development Environment, Assembler and Compiler, Linkers and Debuggers, ARM, Thumb & Thumb2 instructions.

Cortex M4 Microcontrollers & Peripherals

To be competent, the user/individual on the job must be able to:

PC8. Understand the Cortex M4-based controller architecture, Memory mapping and Cortex M4 Peripherals.

PC9. Understand the Cortex M4 interrupt handling –NVIC

PC10. Application development with Cortex M4 controllers using standard peripheral libraries.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Construct Embedded C programming logic
KU2. Analyze and optimize the performance of ARM Controller

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Maintain work-related notes and records
GS2. Read the relevant literature to get the latest updates about the field of work

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
'C' and Embedded C	**17**	**15**	**6**	**5**
PC1. Understand the fundamentals of C programming	-	-	-	-
PC2. Programming with embedded C in contrast with C language	-	-	-	-
PC3. Develop applications using Embedded C Programming	-	-	-	-
Embedded Concepts	**17**	**15**	**6**	**5**
PC4. Understand the various hardware and software architecture of embedded systems	-	-	-	-
PC5. Hands-on exercises on development and debugging tools	-	-	-	-
Introduction to ARM Cortex Architecture	**18**	**15**	**6**	**5**
PC6. Understand The ARM Architecture, Overview of Cortex Architecture, Register Set and Modes	-	-	-	-
PC7. Hands-on exposure to ARM Cortex M4 Development Environment, Assembler and Compiler, Linkers and Debuggers, ARM, Thumb & Thumb2 instructions.	-	-	-	-
Cortex M4 Microcontrollers & Peripherals	**18**	**15**	**7**	**5**
PC8. Understand the Cortex M4-based controller architecture, Memory mapping and Cortex M4 Peripherals.	-	-	-	-
PC9. Understand the Cortex M4 interrupt handling –NVIC	-	-	-	-
PC10. Application development with Cortex M4 controllers using standard peripheral libraries.	-	-	-	-
NOS Total	**70**	**60**	**25**	**20**

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0201
NOS Name	Embedded System Design with Embedded C Programming on ARM Cortex Micro-controllers
Sector	Electronics
Sub-Sector	Sensor, Instrumentation and Cyber Physical System
Occupation	Embedded System Design, IoT Solution
NSQF Level	6
Credits	7
Version	1.0
Last Reviewed Date	23/06/2023
Next Review Date	23/06/2026
NSQC Clearance Date	23/06/2023

NIE/ELE/N0202: Implementation of Complex Embedded Systems using suitable Operating Systems

Description

Students will acquire proficiency in configuring, deploying, and debugging Linux OS on embedded systems through a structured curriculum. Starting with a solid grasp of Linux fundamentals and kernel architecture, they'll set up cross-compilation tools for their chosen target board. Practical knowledge will be gained in kernel and boot loader configuration, root filesystem creation, and device tree implementation.

Scope

The scope covers the following :

- To Skilling the students in Configure, Deploying and Debugging the Linux OS onto a Target Board to build a complete Embedded Product using Linux Kernel.

Elements and Performance Criteria

Basic Operating System Concepts

To be competent, the user/individual on the job must be able to:

PC1. Understand the Fundamentals of Embedded Linux OS,Applications and Products.

PC2. Configure Linux environment for ARM based Target Boards.

Tool-chain: Configuration and Cross-Compilation

To be competent, the user/individual on the job must be able to:

PC3. Understand the Embedded Environment Tools, GNU Tool-chain Cross Compilers.

PC4. Configure Tool-Chain for ARM Platforms.

Embedded Linux Kernel

To be competent, the user/individual on the job must be able to:

PC5. Embedded Environment Tools, GNU Tool-chain Cross Compilers.

PC6. Demonstrate Linux Booting Process and to configure Linux Kernels on ARM based Embedded Boards.

Embedded Linux Application Programming

To be competent, the user/individual on the job must be able to:

PC7. Understanding the Kernel Compilation, Booting the kernel using u-boot, Porting Linux in ARM Boar

PC8. Develop ARM based Embedded Applications with Linux OS.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Understand the Linux OS fundamental

KU2. Able to deploy the tool chain in the ARM controller

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Maintain work-related notes and records

GS2. Read the relevant literature to get the latest updates about the field of work

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Basic Operating System Concepts	**8**	**4**	**2**	**1**
PC1. Understand the Fundamentals of Embedded Linux OS,Applications and Products.	-	-	-	-
PC2. Configure Linux environment for ARM based Target Boards.	-	-	-	-
Tool-chain: Configuration and Cross-Compilation	**8**	**4**	**2**	**1**
PC3. Understand the Embedded Environment Tools, GNU Tool-chain Cross Compilers.	-	-	-	-
PC4. Configure Tool-Chain for ARM Platforms.	-	-	-	-
Embedded Linux Kernel	**8**	**4**	**3**	**1**
PC5. Embedded Environment Tools, GNU Tool-chain Cross Compilers.	-	-	-	-
PC6. Demonstrate Linux Booting Process and to configure Linux Kernels on ARM based Embedded Boards.	-	-	-	-
Embedded Linux Application Programming	**6**	**3**	**3**	**2**
PC7. Understanding the Kernel Compilation, Booting the kernel using u-boot, Porting Linux in ARM Boar	-	-	-	-
PC8. Develop ARM based Embedded Applications with Linux OS.	-	-	-	-
NOS Total	**30**	**15**	**10**	**5**

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0202
NOS Name	Implementation of Complex Embedded Systems using suitable Operating Systems
Sector	Electronics
Sub-Sector	Sensor, Instrumentation and Cyber Physical System
Occupation	Embedded System Design, IoT Solution
NSQF Level	6
Credits	3
Version	1.0
Last Reviewed Date	23/06/2023
Next Review Date	23/06/2026
NSQC Clearance Date	23/06/2023

NIE/ELE/N0203: Real-Time Embedded System Design with RTOS

Description

Unraveling the intricacies of Real-Time Operating Systems (RTOS), this hands-on learning experience combines FreeRTOS and STM32 microcontrollers to demystify RTOS concepts practically. Students immerse themselves in creating applications that employ tasks, semaphores, queues, event flags, and timers.

Scope

The scope covers the following :

- To demystifying RTOS concept practically using Free RTOS and STM32 MCUs for tasks, semaphores, queues, event flags and timers based applications.

Elements and Performance Criteria

Understanding of RTOS concepts

To be competent, the user/individual on the job must be able to:

PC1. Understanding of RTOS concepts

PC2. List Step by step method to run RTOS on STM32 MCUs

RTOS internals

To be competent, the user/individual on the job must be able to:

PC3. Use cases for tasks, semaphores, queues, event flags and timers

PC4. Demonstrate RTOS Scheduler with memory Management.

Embedded system using RTOS

To be competent, the user/individual on the job must be able to:

PC5. Design concepts needed to build an embedded system using RTOS

PC6. Understand complete ARM Cortex M and FreeRTOS Priority model and its configuration related information’s.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Able to demonstrate an understanding of RTOS fundamentals

KU2. Able to perform the hands-on using free RTOS tool

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Maintain work-related notes and records

GS2. Read the relevant literature to get the latest updates about the field of work

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Understanding of RTOS concepts	**15**	**8**	**4**	**5**
PC1. Understanding of RTOS concepts	-	-	-	-
PC2. List Step by step method to run RTOS on STM32 MCUs	-	-	-	-
RTOS internals	**11**	**5**	**3**	**3**
PC3. Use cases for tasks, semaphores, queues, event flags and timers	-	-	-	-
PC4. Demonstrate RTOS Scheduler with memory Management.	-	-	-	-
Embedded system using RTOS	**11**	**5**	**4**	**4**
PC5. Design concepts needed to build an embedded system using RTOS	-	-	-	-
PC6. Understand complete ARM Cortex M and FreeRTOS Priority model and its configuration related information's.	-	-	-	-
NOS Total	**37**	**18**	**11**	**12**

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0203
NOS Name	Real-Time Embedded System Design with RTOS
Sector	Electronics
Sub-Sector	Sensor, Instrumentation and Cyber Physical System
Occupation	Embedded System Design, IoT Solution
NSQF Level	6
Credits	3
Version	1.0
Last Reviewed Date	23/06/2023
Next Review Date	23/06/2026
NSQC Clearance Date	23/06/2023

NIE/ELE/N0204: IoT Application Development with Python and Data Analytics

Description

Armed with hands-on experience using development boards and Python, they will seamlessly integrate Machine Learning (ML) algorithms into their projects. This involves connecting sensors in a network, processing data, and applying ML for enhanced functionality. The program empowers students to navigate the intricate realm of IoT, showcasing their ability to leverage Python effectively. With a focus on practical implementation, graduates are well-prepared to contribute to the dynamic landscape of IoT applications, and equipped to excel in the intersection of sensor networks, ML, and development board technology.

Scope

The scope covers the following :

- After successful completion of the NoS, the students will be able to Implement an IoT application with Development Boards and ML algorithms using Python.

Elements and Performance Criteria

Python Programming

To be competent, the user/individual on the job must be able to:

PC1. Understand the concept of object oriented programming and its applications

PC2. Develop problem solving capability using python scripts

IoT Concepts

To be competent, the user/individual on the job must be able to:

PC3. Understand the evolution of IoT, reference architecture, building block and challenges

PC4. Implement an IoT application using Development Boards

Data Science and Analytics

To be competent, the user/individual on the job must be able to:

PC5. Choose right Data Analytic/ Machine learning tool for various IoT application

PC6. Able to use Data Visualization tools for interactive dynamic visualizations

Statistical and Machine Learning

To be competent, the user/individual on the job must be able to:

PC7. Able to Understand the mathematical principles required for Data Analytics and Machine Learning.

PC8. Able to implement Various ML algorithms using Python.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Gain the expertise in Python script

KU2. Able to analyze the data from the sensor fleet network
KU3. expertise in prediction at the cloud and edge devices

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Maintain work-related notes and records
GS2. Read the relevant literature to get the latest updates about the field of work
GS3. Communicate politely and professionally

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Python Programming	**25**	**17**	**7**	**5**
PC1. Understand the concept of object oriented programming and its applications	-	-	-	-
PC2. Develop problem solving capability using python scripts	-	-	-	-
IoT Concepts	**25**	**17**	**7**	**5**
PC3. Understand the evolution of IoT, reference architecture, building block and challenges	-	-	-	-
PC4. Implement an IoT application using Development Boards	-	-	-	-
Data Science and Analytics	**25**	**12**	**8**	**5**
PC5. Choose right Data Analytic/ Machine learning tool for various IoT application	-	-	-	-
PC6. Able to use Data Visualization tools for interactive dynamic visualizations	-	-	-	-
Statistical and Machine Learning	**25**	**11**	**8**	**5**
PC7. Able to Understand the mathematical principles required for Data Analytics and Machine Learning.	-	-	-	-
PC8. Able to implement Various ML algorithms using Python.	-	-	-	-
NOS Total	**100**	**57**	**30**	**20**

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0204
NOS Name	IoT Application Development with Python and Data Analytics
Sector	Electronics
Sub-Sector	Sensor, Instrumentation and Cyber Physical System
Occupation	Embedded System Design, IoT Solution
NSQF Level	6
Credits	9
Version	1.0
Last Reviewed Date	23/06/2023
Next Review Date	23/06/2026
NSQC Clearance Date	23/06/2023

NIE/ELE/N0205: Complex Embedded Application Development with an understanding of Embedded Protocols and Device Driver Concepts

Description

This program aims to provide students with vital insights into embedded protocols and the practical expertise needed to implement device drivers within the Linux kernel. Covering essential aspects of embedded systems, the curriculum encompasses protocol specifications and in-depth exploration of device driver architecture.

Scope

The scope covers the following :

- To equip the students with the information required in embedded protocols and to implement the device drivers in the Linux kernel.

Elements and Performance Criteria

Embedded Protocols

To be competent, the user/individual on the job must be able to:

PC1. Demonstrate Different embedded protocols like SPI, I2C, USB and CAN.

PC2. Able to choose right protocol for the different embedded applications.

Linux Device drivers

To be competent, the user/individual on the job must be able to:

PC3. Build driver program for various devices in Linux kernel.

PC4. Hands-on expertise in File Operations in Char Device

Kernel driver for external peripherals

To be competent, the user/individual on the job must be able to:

PC5. Able to develop Kernal driver for Keyboard, Speaker, and RTC

PC6. Understand the Communication between Hardware Memory Space and IO

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Practical understanding of the various serial protocols

KU2. Able to create linux device drivers for system peripherals

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Maintain work-related notes and records

GS2. Read the relevant literature to get the latest updates about the field of work

GS3. Communicate politely and professionally

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Embedded Protocols	**22**	**11**	**8**	**2**
PC1. Demonstrate Different embedded protocols like SPI, I2C, USB and CAN.	-	-	-	-
PC2. Able to choose right protocol for the different embedded applications.	-	-	-	-
Linux Device drivers	**22**	**11**	**8**	**2**
PC3. Build driver program for various devices in Linux kernel.	-	-	-	-
PC4. Hands-on expertise in File Operations in Char Device	-	-	-	-
Kernel driver for external peripherals	**23**	**11**	**9**	**1**
PC5. Able to develop Kernal driver for Keyboard, Speaker, and RTC	-	-	-	-
PC6. Understand the Communication between Hardware Memory Space and IO	-	-	-	-
NOS Total	**67**	**33**	**25**	**5**

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0205
NOS Name	Complex Embedded Application Development with an understanding of Embedded Protocols and Device Driver Concepts
Sector	Electronics
Sub-Sector	Sensor, Instrumentation and Cyber Physical System
Occupation	Embedded System Design, IoT Solution
NSQF Level	6
Credits	6
Version	1.0
Last Reviewed Date	23/06/2023
Next Review Date	23/06/2026
NSQC Clearance Date	23/06/2023

DGT/VSQ/N0104: Employability Skills (120 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the current job market requirement and future of work

PC2. identify and explore learning and employability relevant portals

PC3. research about the different industries, job market trends, latest skills required and the available opportunities

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC4. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. for personal growth and the nation's progress

PC5. follow personal values and ethics such as honesty, integrity, caring and respecting others, etc.

PC6. follow and promote environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC7. recognize the significance of 21st Century Skills for employment

PC8. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

PC9. adopt a continuous learning mindset for personal and professional development

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC10. use English as a medium of formal and informal communication while dealing with topics of everyday conversation in different contexts

PC11. speak over the phone in English, in an audible manner, using appropriate greetings, opening, and closing statements both on personal and work front

PC12. read and understand routine information, notes, instructions, mails, letters etc. written in English

PC13. write short messages, notes, letters, e-mails etc., using accurate English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

PC14. identify career goals based on the skills, interests, knowledge, and personal attributes

PC15. prepare a career development plan with short- and long-term goals

Communication Skills

To be competent, the user/individual on the job must be able to:

PC16. follow verbal and non-verbal communication etiquette while communicating in professional and public settings

PC17. use active listening techniques for effective communication

PC18. communicate in writing using appropriate style and format based on formal or informal requirements

PC19. work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC20. • ensure personal behaviour, conduct, and use appropriate communication by taking gender into • consideration

PC21. empathize with a PwD and aid a PwD, if asked

PC22. escalate any issues related to sexual harassment at the workplace in accordance with the POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC23. identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.

PC24. carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook

PC25. identify common components of salary and compute income, expenses, taxes, investments etc

PC26. identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC27. operate digital devices and use their features and applications securely and safely

PC28. carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.

PC29. display responsible online behaviour while using various social media platforms

PC30. create a personal email account, send and process received messages as per requirement

PC31. carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications

PC32. utilize virtual collaboration tools to work effectively

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC33. identify different types of Entrepreneurship and Enterprises

PC34. use research and networking skills to identify and assess opportunities for potential business

PC35. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion

PC36. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

PC37. identify different types of customers

PC38. identify and respond to customer requests and needs in a professional manner

PC39. use appropriate tools to collect customer feedback

PC40. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC41. create a professional Curriculum vitae (Résumé)

PC42. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC43. apply to identified job openings using offline /online methods as per requirement

PC44. answer questions politely, with clarity and confidence, during recruitment and selection

PC45. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. different learning and employability related portals

KU3. various constitutional and personal values

KU4. different environmentally sustainable practices and their importance

KU5. Twenty first (21st) century skills and their importance

KU6. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU7. importance of career development and setting long- and short-term goals

KU8. Do's and don'ts of effective communication

KU9. POSH Act

KU10. inclusivity and its importance

KU11. different types of disabilities and appropriate verbal and non-verbal communication and behaviour towards PwD

KU12. different types of financial institutes, products, and services

KU13. components of salary and how to compute income and expenditure

KU14. importance of maintaining safety and security in offline and online financial transactions

KU15. different legal rights and laws

KU16. different types of digital devices and the procedure to operate them safely and securely

KU17. how to create and operate an e- mail account

KU18. use applications such as word processors, spreadsheets etc.

KU19. different types of Enterprises and ways to identify business opportunities

KU20. types and needs of customers

KU21. how to apply for a job and prepare for an interview

KU22. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence in English and other languages

GS2. communicate effectively using appropriate language in formal and informal settings

GS3. behave politely and appropriately with all to maintain effective work relationship

GS4. how to work in a virtual mode, using various technological platforms

GS5. perform calculations efficiently

GS6. solve problems effectively

GS7. pay attention to details

GS8. manage time efficiently

GS9. maintain hygiene and sanitization to avoid infection

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
Introduction to Employability Skills	**1**	**1**	**-**	**-**
PC1. understand the significance of employability skills in meeting the current job market requirement and future of work	-	-	-	-
PC2. identify and explore learning and employability relevant portals	-	-	-	-
PC3. research about the different industries, job market trends, latest skills required and the available opportunities	-	-	-	-
Constitutional values – Citizenship	**2**	**1**	**-**	**-**
PC4. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. for personal growth and the nation's progress	-	-	-	-
PC5. follow personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC6. follow and promote environmentally sustainable practices	-	-	-	-
Becoming a Professional in the 21st Century	**2**	**3**	**-**	**-**
PC7. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC8. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
PC9. adopt a continuous learning mindset for personal and professional development	-	-	-	-
Basic English Skills	**2**	**3**	**-**	**-**

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. use English as a medium of formal and informal communication while dealing with topics of everyday conversation in different contexts	-	-	-	-
PC11. speak over the phone in English, in an audible manner, using appropriate greetings, opening, and closing statements both on personal and work front	-	-	-	-
PC12. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC13. write short messages, notes, letters, e-mails etc., using accurate English	-	-	-	-
Career Development & Goal Setting	**1**	**2**	**-**	**-**
PC14. identify career goals based on the skills, interests, knowledge, and personal attributes	-	-	-	-
PC15. prepare a career development plan with short- and long-term goals	-	-	-	-
Communication Skills	**2**	**3**	**-**	**-**
PC16. follow verbal and non-verbal communication etiquette while communicating in professional and public settings	-	-	-	-
PC17. use active listening techniques for effective communication	-	-	-	-
PC18. communicate in writing using appropriate style and format based on formal or informal requirements	-	-	-	-
PC19. work collaboratively with others in a team	-	-	-	-
Diversity & Inclusion	**1**	**2**	**-**	**-**
PC20. • ensure personal behaviour, conduct, and use appropriate communication by taking gender into • consideration	-	-	-	-
PC21. empathize with a PwD and aid a PwD, if asked	-	-	-	-

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. escalate any issues related to sexual harassment at the workplace in accordance with the POSH Act	-	-	-	-
Financial and Legal Literacy	**2**	**3**	**-**	**-**
PC23. identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.	-	-	-	-
PC24. carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook	-	-	-	-
PC25. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC26. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
Essential Digital Skills	**2**	**3**	**-**	**-**
PC27. operate digital devices and use their features and applications securely and safely	-	-	-	-
PC28. carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.	-	-	-	-
PC29. display responsible online behaviour while using various social media platforms	-	-	-	-
PC30. create a personal email account, send and process received messages as per requirement	-	-	-	-
PC31. carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications	-	-	-	-
PC32. utilize virtual collaboration tools to work effectively	-	-	-	-
Entrepreneurship	**2**	**3**	**-**	**-**

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC33. identify different types of Entrepreneurship and Enterprises	-	-	-	-
PC34. use research and networking skills to identify and assess opportunities for potential business	-	-	-	-
PC35. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC36. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
Customer Service	**1**	**2**	**-**	**-**
PC37. identify different types of customers	-	-	-	-
PC38. identify and respond to customer requests and needs in a professional manner	-	-	-	-
PC39. use appropriate tools to collect customer feedback	-	-	-	-
PC40. follow appropriate hygiene and grooming standards	-	-	-	-
Getting ready for apprenticeship & Jobs	**2**	**4**	**-**	**-**
PC41. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC42. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC43. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC44. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC45. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	**20**	**30**	**-**	**-**

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0104
NOS Name	Employability Skills (120 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	6
Credits	4
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

• Assessment of the qualification evaluates candidates to ascertain that they can integrate knowledge, skills and values for carrying out relevant tasks as per the

• defined learning outcomes and assessment criteria.

• The underlying principle of assessment is fairness and transparency. The evidence of the outcomes and assessment criteria. competence acquired by the candidate

• can be obtained by conducting Theory (Online), Practical assessment, Internal assessment, Project/Presentation/ Assignment, Major Project. The emphasis is on the

• practical demonstration of skills & knowledge gained by the candidate through the training. Each OUTCOME is assessed & marked separately. A candidate is

• required to pass all OUTCOMES individually based on the passing criteria.

• About Examination Pattern:

• 1. The question papers for the theory and practical exams are set by the Examination wing (assessor) of

NIELIT HQS.

• 2. The assessor assigns roll number

• 3. The assessor carries out theory online assessments through remote proctoring methodology. Theory examination would be conducted online and the paper comprise of MCQ. Conduct of assessment are through trained proctors. Once the test begins, remote proctors have full access to candidate's video feeds and computer screens. Proctors authenticate the candidate based on registration details, pre-test image captured and I- card in possession of the candidate. Proctors can chat with candidates or give warnings to candidates. Proctors can also take screenshots, terminate a specific user's test session, or re-authenticate candidates based on video feeds.

• 4. An External Examiner/ Observer may be deployed including NIELIT officials for evaluation of Practical examination/ internal assessment / Project/ Presentation/. Major Project (if applicable) would be evaluated preferably by external/ subject expert including NIELIT officials.

• 5. Pass percentage would be 50% marks in each component.

• 6. Candidates may apply for re-examination within the validity of registration (only in the assessment component in which the candidate failed).

• 7. For re-examination prescribed examination fee is required to be paid by the candidate only for the assessment component in which the candidate wants to reappear.

• 8. There would be no exemption for any paper/module for candidates having similar qualifications or skills.

• 9. The examination will be conducted in English language only.

• Quality assurance activities: A pool of questions is created by a subject matter expert and moderated by other SME. Test rules are set beforehand. Random set of questions which are according to syllabus appears which may differ from candidate to candidate. Confidentiality and impartiality are maintained during all the examination and evaluation processes.

Minimum Aggregate Passing % at QP Level : 50

(**Please note**: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
NIE/ELE/N0201.Embedded System Design with Embedded C Programming on ARM Cortex Micro-controllers	70	60	25	20	175	25
NIE/ELE/N0202.Implementation of Complex Embedded Systems using suitable Operating Systems	30	15	10	5	60	9
NIE/ELE/N0203.Real-Time Embedded System Design with RTOS	37	18	11	12	78	11
NIE/ELE/N0204.IoT Application Development with Python and Data Analytics	100	57	30	20	207	30
NIE/ELE/N0205.Complex Embedded Application Development with an understanding of Embedded Protocols and Device Driver Concepts	67	33	25	5	130	18
DGT/VSQ/N0104.Employability Skills (120 Hours)	20	30	-	-	50	7
Total	**324**	**213**	**101**	**62**	**700**	**100**

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.